

NETSHIELD: Timely Evidence Feasibility under Shared Bottlenecks for Multi-Probe Edge Intrusion Detection

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Abstract—Multi-probe edge security systems generate network demand endogenously: a runtime action determines how much evidence is offered to a shared transport-and-core service. We develop NETSHIELD as a feasibility framework for this coupling. For n probes, let m retain remote evidence in a service round, of which k send full evidence of size B and the remaining $m - k$ send summaries of size $S < B$. One-round byte feasibility is exactly $kB + (m - k)S \leq C$. If non-selected probes may send no remote evidence, the maximum one-round full-evidence count is $K_{\text{full}}^{\max} = \min\{n, \lfloor C/B \rfloor\}$; requiring every probe to remain remotely represented reduces this to a summary-preserving boundary. For the three-probe testbed with $B = 128$ KiB and $S = 16$ KiB, universal per-round coverage costs one full-evidence slot at both low and mid capacity, but none at high capacity. We place this result inside a hierarchy of same-round, finite-deadline, and long-run load feasibility.

The frozen independent queue-penalized controller exhibits a symmetry-induced full/none limit cycle. Under a reduced saturated regime, flow conservation gives a long-run full-request fraction $p^* = \min(1, C/(nB))$; a 20-seed phase-randomized software study matches 1/3, 2/3, and 1 at the three capacities. A separately frozen 36-arm cyclic rotating full-admission (CRFA) hardware extension then tests the unrestricted frontier directly. Its post-release timely full-core output is 0.3056, 0.6327, and 0.9660 per generated event at low, mid, and high capacity, corresponding to 91.7%, 94.9%, and 96.6% of the byte frontier. At mid capacity, relaxing universal per-round coverage raises timely full-core output from 0.2938 under coordinated summary-preserving admission to 0.6327 under CRFA (paired difference +0.3389, 95% session-bootstrap interval [0.3306, 0.3478]). At high capacity, where the coverage constraint is non-binding, the paired difference is -0.0037 [$-0.0133, 0.0046$]. Every selected CRFA full result is eventually returned, so the remaining gap to the byte frontier is a physical timing residual rather than loss of evidence.

The primary 216-arm campaign and the 36-arm extension together account for 252 accepted hardware arms and 68,040 measured events. Their independently reconstructed service traces contain 74,340 rounds with zero shared application-byte cap violations. The resulting contribution is a networking characterization of the price of per-round evidence coverage, controller synchronization, and the gap between byte feasibility and physical deadline attainment.

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I. INTRODUCTION

Edge intrusion detection increasingly combines local inference with remote evidence handling. A probe may retain a decision locally, emit a compact summary, or upload richer evidence for core-side execution and correlation. These actions differ not only in security utility but also in the number of bytes injected into a shared service. Evidence generation is therefore an *endogenous networking decision*: the controller chooses the offered load before the bottleneck serves it.

This distinction matters when several probes act concurrently. If three probes each request a 128 KiB full upload, the aggregate request is 393,216 B. Under a service budget smaller than that amount, every local request may be reasonable in isolation while the joint tuple is infeasible. The relevant question is not merely whether the detector runs, but how much remote-evidence coverage and full-evidence richness can be delivered within a response horizon.

Recent work on remote inference explicitly couples communication timing, feature freshness, and inference quality [25], [26], while modern edge systems expose request scheduling and admission as first-class resource-allocation problems [12]–[14]. Security systems likewise show that collecting and transporting the data needed for detection can itself become a bottleneck [8], [9]. NETSHIELD connects these observations in a multi-probe security setting by making the evidence action itself part of the network load model.

The formal controller artifacts also expose a second networking issue: synchronization. The frozen independent queue-aware selector uses

$$\psi_i(a, t) = vu(a) - Q_i(t)b(a)/10^6, \quad (1)$$

where v weights the frozen utility $u(a)$ against the queue-payload penalty, $Q_i(t)$ is the local evidence queue in bytes, and $b(a)$ is the action payload in bytes. When identical probes decide at synchronized epochs, equal queue states feed back into equal actions. At mid capacity, this produces a period-two full/none orbit. A poor synchronized result therefore cannot be interpreted as evidence that decentralization itself is inferior.

The paper instead asks four questions:

- **RQ1—Feasibility and the price of coverage:** What full-evidence throughput is byte-feasible, and what is lost when every probe must retain remote evidence in every round?
- **RQ2—Controller realization and synchronization:** How do coordinated and independent controllers occupy the feasible region, and when does local queue feedback create synchronized limit cycles?
- **RQ3—Physical timing residual:** Once a byte allocation is feasible, how do orchestration and wireless timing affect post-release and schedule-inclusive service attainment?
- **RQ4—Scaling and reproducibility:** Do the analytical boundaries hold as n changes, and can the hardware result chain be reconstructed from occurrence-level evidence?

The paper makes four contributions. First, the exact coverage-conditioned region and unrestricted frontier $K_{\text{full}}^{\text{max}} = \min\{n, \lfloor C/B \rfloor\}$ quantify the *price of universal per-round remote coverage*. A cyclic construction equalizes frontier-attaining full-evidence opportunities over time.

Second, a three-layer characterization separates same-round admission, finite- D deadline feasibility, and long-run load admissibility, distinguishing instantaneous constraints from deadline service and average-rate conditions.

Third, the analysis explains the synchronized independent limit cycle and gives a conditional rate-balance law for the desynchronized H-FULL/H-NONE regime. Phase-randomized software conforms to its predicted request fractions, isolating synchronization from decentralization.

Fourth, three heterogeneous physical Raspberry Pi probes test both boundaries. A 216-arm primary campaign characterizes coordinated and independent control; a separately frozen 36-arm CRFA extension instantiates the unrestricted 1/2/3-full operating points. It attains 91.7%, 94.9%, and 96.6% of the low/mid/high byte frontier under the 1-s post-release SLO. Auditable hardware timestamps separate byte feasibility from orchestration and wireless/core timing residuals in both campaigns.

a) *Claim boundary.*: NETSHIELD does not claim a new intrusion classifier, state-of-the-art CICIOT2023 accuracy [5], global stochastic-control optimality, or empirical superiority of the safety filter. The primary claims concern network-byte feasibility, the cost of per-round coverage, controller synchronization, and service timing.

II. RELATED WORK

A. IoT Intrusion Detection and Generalization

Lightweight IoT IDS research has produced compact anomaly detectors, device fingerprints, and online pipelines, including SVELTE, IoT Sentinel, N-BaIoT, and Kitsune [1]–[4]. More recent work addresses lightweight open-set detection for previously unseen IIoT attacks [7]. These systems primarily optimize the detector. NETSHIELD instead freezes the detector/evidence action set and studies whether the resulting evidence requests can be sustained jointly. The distinction is important because closed-set replay accuracy does not by itself establish deployment performance outside the training/test world [6].

B. Security Data Collection and Evidence Overhead

Recent security systems make the cost of acquiring evidence explicit. ISDC prioritizes network-defense data collection under constrained in-network resources [8], while ChainPatrol trades attack visibility against traffic and delay overhead for service-function-chain verification [9]. These works motivate treating security evidence as a resource-consuming network object rather than a free by-product of detection. Our control point is complementary: the runtime action determines both how many probes remain remotely represented and how rich their evidence is before shared-service admission.

C. Edge Scheduling and Admission Control

Mobile/edge computing studies placement and offloading under latency, bandwidth, and energy constraints [10], [11]. Recent systems make scheduling and admission more explicit: Cur-CoEdge coordinates edge-cloud request scheduling [12], EdgeTimer adapts scheduling timescales in MEC [13], and OACR² jointly controls online admission and resource reservation for network slices [14]. Classical fair queueing and GPS instead determine how a bottleneck serves traffic after it has entered queues [15], [16]. NETSHIELD adds an upstream decision in which the evidence action itself determines offered load.

D. Queue-Aware Network Control

Queue-aware scheduling and Lyapunov drift-plus-penalty provide a foundation for balancing utility against backlog [17], [18]. Recent ToN work also combines Lyapunov-style queue scheduling with online learning under non-stationary conditions [19]. The frozen selector studied here is deliberately narrower: it is the finite action score in (1), not a claimed implementation of a complete stochastic Lyapunov optimizer. We therefore use the descriptive names independent queue-penalized greedy (I-QPG), coordinated QPG (C-QPG), and safety-filtered coordinated QPG (S-CQPG).

E. Timely Remote Inference and Serving

Prediction-serving systems expose latency-resource trade-offs through batching, provisioning, and execution control. Clipper, Nexus, Clockwork, and InferLine are representative systems [20]–[23]; SuperServe extends this line to fine-grained serving under unpredictable workloads [24]. Networking work on remote inference instead optimizes feature freshness, packet length, or transmission timing [25], [26]. NETSHIELD controls an upstream security decision: the action determines the quantity and richness of evidence offered to the network before the transport/core service executes.

F. Runtime Safety Filters and Prior Work

Runtime shields and predictive safety filters separate policy choice from hard enforcement [27], [28]. We retain that architecture as an optional local-feasibility interface, but the present campaign does not show an independent performance advantage over ordinary capacity admission.

TABLE I
FROZEN ACTION PAYLOADS AND UTILITY SCORES.

Action	Payload (B)	Utility
L-NONE	0	0.758501
L-ALERT	256	0.825168
L-SUMMARY	4,096	0.891835
M-NONE	0	0.766968
M-ALERT	256	0.833635
M-SUMMARY	8,192	0.900302
H-NONE	0	0.768378
H-ALERT	256	0.835044
H-SUMMARY	16,384	0.901711
H-FULL	131,072	0.968378

Our prior ACISP 2026 Systematization of Knowledge surveyed telemetry-aware runtime assurance for always-on on-device IDS and presented TQS-IDS as a design template [29]. It did not implement the shared- bottleneck formulation, the queue-penalized controllers, the event contract, or the three-Pi hardware campaign studied here.

III. SYSTEM MODEL AND FEASIBILITY ANALYSIS

A. Shared Evidence Service

Consider n edge probes sharing an application-level service budget $C(t)$ bytes in coordination round t . Probe i has evidence queue $Q_i(t)$ and chooses an admitted action $a_i(t)$ with payload $b(a_i)$. The common service grants $g_i(t)$ satisfy

$$g_i(t) \geq 0, \quad \sum_{i=1}^n g_i(t) \leq C(t), \quad (2)$$

and the queue evolves as

$$Q_i(t+1) = [Q_i(t) + b(a_i(t)) - g_i(t)]^+. \quad (3)$$

The low, mid, and high labels used in the hardware campaign denote controlled application-byte budgets per coordination round, not measured Wi-Fi PHY throughput. Throughout, MB denotes 10^6 bytes, while KiB and MiB denote 2^{10} and 2^{20} bytes, respectively.

B. Frozen Action Lattice

The implementation exposes ten frozen actions spanning light, medium, and heavy detector execution and evidence levels from no upload to full core-side evidence. Table I lists the frozen payload and utility fields.

The coordinated regimes eventually operate on the H-SUMMARY–H-FULL edge of this lattice, which admits an exact coverage–richness boundary.

C. One-Round Coverage–Richness Region

Let $m \leq n$ probes retain remotely delivered evidence in a round. Among them, let $k \leq m$ send full evidence of size B ; the other $m - k$ send summaries of size S , where $0 < S < B$. The remaining $n - m$ probes generate no remote evidence.

Theorem 1 (Coverage-conditioned one-round feasibility). *A (k, m) allocation is byte-feasible within one idealized service round if and only if*

$$kB + (m - k)S \leq C. \quad (4)$$

For fixed m , if $C \geq mS$, the maximum feasible full-evidence count is

$$k^*(C, m) = \min \left\{ m, \left\lfloor \frac{C - mS}{B - S} \right\rfloor \right\}. \quad (5)$$

If $C < mS$, no allocation preserving remote evidence for all m probes is feasible.

Proof. The payload is $mS + k(B - S)$. If $C < mS$, even the all-summary allocation is infeasible. Otherwise, since $B - S > 0$, feasibility is equivalent to $k(B - S) \leq C - mS$; the largest feasible integer k is therefore (5), clipped by m . $\square$

Define $\rho = C/(nB)$, $\sigma = S/B$, $r = m/n$, and $f = k/n$. Ignoring finite- n granularity, (4) becomes

$$\sigma r + (1 - \sigma)f \leq \rho. \quad (6)$$

This continuous relaxation separates the capacity spent on remote coverage from that spent upgrading summaries to full evidence. The integer boundary fixes the feasible set independently of the controller: the subsequent analysis asks which point a controller selects and how much of that byte-feasible allocation meets the physical service horizon.

Corollary 1 (Full-evidence byte frontier and price of coverage). *If non-selected probes may send no remote evidence, the maximum number of full-evidence probes whose bytes can fit in one idealized round is*

$$K_{\text{full}}^{\max}(C) = \min \left\{ n, \left\lfloor \frac{C}{B} \right\rfloor \right\}. \quad (7)$$

If every probe must remain remotely represented, the maximum is

$$K_{\text{all}}^{\max}(C) = \left[\left\lfloor \frac{C - nS}{B - S} \right\rfloor \right]_0^n, \quad (8)$$

provided $C \geq nS$. The full-slot price of universal per-round remote coverage is

$$\Pi_{\text{cov}}(C) = K_{\text{full}}^{\max}(C) - K_{\text{all}}^{\max}(C). \quad (9)$$

Proof. For any feasible pair, $m \geq k$ and $S > 0$. For fixed k , payload is minimized by $m = k$, which costs kB . Thus no allocation can contain more than $\lfloor C/B \rfloor$ full requests, and choosing $m = k$ attains that count. Equation (8) is Theorem 1 with $m = n$. $\square$

The quantity $K_{\text{full}}^{\max}/n$ is a one-round *byte-completion* frontier. It is not by itself a physical 1-s result-return guarantee because core processing, wireless timing, and orchestration occur after or alongside byte service.

D. Three Feasibility Layers

The one-round region is the strongest of three useful service notions. Let A_t denote total admitted bytes released at the beginning of round t , and consider an ideal preemptive fluid server of capacity C bytes per round.

Proposition 1 (Same-round, finite-deadline, and load feasibility). *For a common relative deadline of $D \geq 1$ rounds,*

a necessary and sufficient demand condition under the ideal fluid/EDF model is

$$\sum_{\tau=s}^t A_\tau \leq C(t - s + D), \quad \forall s \leq t. \quad (10)$$

When $D = 1$, this reduces to same-round feasibility $A_t \leq C$ for every t . Any finite- D feasible sequence also satisfies the long-run load condition

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T A_t \leq C. \quad (11)$$

Thus, for finite D ,

$$\mathcal{F}_{\text{same}} \subseteq \mathcal{F}_D \subseteq \mathcal{F}_{\text{rate}}. \quad (12)$$

Proof. A job released in round τ has deadline $\tau + D - 1$. The processor-demand criterion for a preemptive single fluid server requires that, for any service interval, the work whose release times and deadlines both fall inside that interval not exceed available service. Re-indexing the interval gives (10). Setting $D = 1$ yields per-round feasibility. Finally, using $s = 1$ and dividing (10) by T gives $T^{-1} \sum_{t=1}^T A_t \leq C[1 + (D - 1)/T]$, whose limsup is (11). $\square$

Equation (11) is a load-admissibility condition, not a claim that an arbitrary burst process with mean exactly C has bounded delay. Standard queue-stability conclusions require additional arrival assumptions and typically strict slack [17], [18].

E. Deterministic Overload Growth

Proposition 2 (Fixed-full overload law). *If every probe generates B bytes per generation round, the initial queues are empty, aggregate service is work conserving, and $C < nB$, then after T generation rounds*

$$Q_\Sigma(T) = T(nB - C). \quad (13)$$

Under symmetric fair service, per-probe backlog is approximately $T(B - C/n)$ up to integer-grant rounding.

Proof. Each round adds nB bytes. Work-conserving service removes $\min\{Q_\Sigma(t) + nB, C\} = C$, since $nB > C$ even when the queue is empty. Thus backlog grows by exactly $nB - C$ each round; summing from the empty initial state yields (13). $\square$

F. Independent Queue-Penalized Selection

The frozen independent policy uses (1). At $v = 1$, a positive-payload action a can beat H-NONE only while

$$Q_i < q_a = \frac{[u(a) - u(\text{H-NONE})]10^6}{b(a)}. \quad (14)$$

For the relevant actions, $q_{\text{H-FULL}} \approx 1.526$ B, $q_{\text{H-SUMMARY}} \approx 8.138$ B, and the largest threshold over any positive-payload action is about 260.4 B (H-ALERT). At mid capacity, a synchronized three-probe full round leaves $q^+ = B - 2B/3 = B/3 \approx 43,691$ B (42.7 KiB) per probe.

Proposition 3 (Symmetry-induced period-two orbit). *Suppose synchronized probes have identical state and empty queues,*

choose H-FULL when $Q_i = 0$, and a synchronized full round leaves residual q^+ . If $q^+ > \max_{a:b(a)>0} q_a$ and a zero-payload round can drain q^+ , then the controller enters the orbit $\text{H-FULL} \rightarrow \text{H-NONE} \rightarrow \text{H-FULL} \rightarrow \dots$.

Proof. After the full round, every positive-payload action scores below H-NONE. The next zero-payload round drains the residual and restores the empty state, so symmetry makes all probes repeat the same sequence. $\square$

Proposition 4 (Rate balance in the reduced desynchronized regime). *Consider the reduced regime in which each probe receives one request opportunity per round and emits either B bytes (H-FULL) or zero bytes (H-NONE). Let p be the long-run fraction of request opportunities that emit B . Aggregate rate stability implies*

$$p \leq \min\left(1, \frac{C}{nB}\right) = \min(1, \rho). \quad (15)$$

If $C < nB$ and the shared server is asymptotically saturated, equality holds. If $C \geq nB$ and an unbacklogged probe requests full evidence at every opportunity, then $p = 1$. Hence the saturated reduced regime has

$$p^* = \min(1, \rho). \quad (16)$$

Proof. Queue conservation over T rounds gives $Q_\Sigma(T) = Q_\Sigma(0) + BnpT - S(T) + o(T)$. Rate stability and the service bound $S(T)/T \leq C + o(1)$ imply $npB \leq C$, together with the trivial $p \leq 1$. When $C < nB$ and $S(T)/T \rightarrow C$, rate stability forces $npB = C$. When $C \geq nB$, requesting at every unbacklogged opportunity gives $p = 1$ and remains load-feasible. $\square$

The proposition is conditional: it characterizes a saturated reduced regime rather than claiming that every decentralized policy must attain ρ .

G. A Fair Construction at the Full-Evidence Frontier

Let $K = K_{\text{full}}^{\max}(C)$. A simple cyclic construction selects exactly K probes for H-FULL in each round and assigns H-NONE to the rest.

Proposition 5 (Cyclic rotating full admission). *Index probes modulo n and define*

$$\mathcal{S}_t = \{(tK + j) \bmod n : j = 0, \dots, K - 1\}. \quad (17)$$

Then every round is one-round byte-feasible and attains $K_{\text{full}}^{\max}$. The selection pattern has period

$$T_{\text{fair}} = \frac{n}{\gcd(n, K)}, \quad (18)$$

and each probe is selected exactly $K/\gcd(n, K)$ times per period.

Proof. Each round injects exactly $KB \leq C$. Advancing the starting index by K returns to the initial residue after $n/\gcd(n, K)$ rounds. Across that period there are KT_{fair} selection slots distributed symmetrically over n residues, giving $K/\gcd(n, K)$ selections per probe. $\square$

TABLE II
APPLICATION-SERVICE BUDGETS.

Regime	Budget (B/round)	Full equivalents
Low	131,072	1
Mid	262,144	2
High	393,216	3

This construction converts per-round coverage into temporal fairness: at mid capacity with $n = 3$ and $K = 2$, the three-round sequence can be $\{P1, P2\} \rightarrow \{P3, P1\} \rightarrow \{P2, P3\}$, so every probe receives two full-evidence opportunities per three rounds. We first verify this schedule under idealized service accounting and then test the same construction in a separately frozen physical extension described in Section IV.

H. Joint Admission and Optional Safety Filtering

C-QPG first obtains per-probe queue-penalized requests and monotonically reduces simultaneous requests until the common budget is satisfied. S-CQPG adds a local safety filter before the same joint admission stage. The filter is not part of the network byte theorems. In the hardware campaign, low-capacity safety filtering and ordinary shared-capacity admission reach the same final summary mix, so no independent performance advantage is claimed.

IV. EXPERIMENTAL METHODOLOGY

A. Physical Testbed and Workload

The testbed uses three physical probes: P1 is a Raspberry Pi 5, P2 a Raspberry Pi 4B, and P3 a Raspberry Pi 3B+. Probe-to-core experiment traffic uses Wi-Fi, and the edge core executes the heavy core path for H-FULL events.

The workload is a frozen feature-vector replay derived from CICIOT2023 [5], not packet replay. Detector models, feature order, action profiles, window plans, and formal configuration are frozen before the primary campaign. Each arm uses five warm-up rounds followed by 90 measured event occurrences per probe, with up to 200 drain rounds. The primary matrix is

$$12 \text{ sessions} \times 6 \text{ methods} \times 3 \text{ capacities} = 216 \text{ arms},$$

giving 648 device-runs and 58,320 measured events. The rotating extension reuses the same three probes, action definitions, session/workload identities, and measured-window contract, but is archived as a separate campaign.

B. Shared Service Budgets

The formal application-byte budgets are shown in Table II. Because $B = 128$ KiB, they correspond to one, two, or three full-evidence byte equivalents. These budgets regulate application payload service and are not measurements of 1/2/3-Mbit/s physical Wi-Fi links.

C. Hardware Methods

Table III summarizes the primary hardware methods.

D. Separately Frozen Rotating-Admission Extension

After the analytical frontier and the primary campaign had been frozen, we executed a separate hardware extension using cyclic rotating full admission (CRFA). In each measured round, CRFA admits exactly $K_{\text{full}}^{\max}(C)$ probes with H-FULL and assigns H-NONE to the remaining probes; admitted identities follow the cyclic schedule in Proposition 5. The extension therefore tests the unrestricted full-evidence frontier rather than another utility optimizer.

The extension uses the same 12 session/workload identities and three service budgets, giving

$$12 \text{ sessions} \times 3 \text{ capacities} = 36 \text{ accepted arms}.$$

The frozen protocol retains the same five warm-up rounds, 90 measured occurrences per probe, and drain rules as the primary campaign. The resulting hardware evidence contains 9,720 measured occurrences and 10,620 reconstructed shared-service rounds.

Across execution, 55 attempts were indexed. Nineteen were excluded in full under pre-specified infrastructure/data-integrity rules, predominantly SSH transport or pre-arm hardware-gate failures; all excluded attempts remain in the attempt index. Timing outcomes were never used as acceptance criteria and partial attempts were not stitched into accepted arms. The 36 accepted arms pass event-identity, action/payload-identity, final-drain, raw-reconstruction, and shared-cap gates, with zero shared-cap violations.

For descriptive comparison with the primary C-QPG results, CRFA arms are matched by the frozen session/workload identifier. Because the extension was executed separately rather than as a concurrent randomized A/B trial, paired session-bootstrap intervals are interpreted as matched descriptive timing contrasts rather than as causal treatment intervals.

E. Event-Level Metric Authority

The occurrence-level event ledger is the scientific source of truth. Each admitted event retains an immutable execution contract specifying whether its result is local or core-dependent and what evidence transfer is required. For a core-required event e , post-release latency is

$$L_{\text{post}}(e) = t_{\text{return}}(e) - t_{\text{created}}(e), \quad (19)$$

and the primary service SLO is $L_{\text{post}} \leq 1$ s. The schedule-inclusive sensitivity instead uses

$$L_{\text{nom}}(e) = t_{\text{return}}(e) - t_{\text{scheduled}}(e). \quad (20)$$

Missing results remain in the denominator and are not counted as on time.

The primary service quantities are core-required fraction, result-return coverage, post-release on-time core-result coverage, conditional on-time rate among core-required events, generated application payload, and queue occupancy. Classifier correctness is not multiplied into the main service metric because correctness is constant on the scored formal replay occurrences.

TABLE III

HARDWARE METHODS. HISTORICAL ARTIFACT IDENTIFIERS ARE RETAINED ONLY IN THE REPRODUCIBILITY PACKAGE, NOT AS ALGORITHMIC CLAIMS.

ID	Paper name	Runtime semantics
A	Fixed local	Always H-NONE; no application evidence upload.
B	Fixed summary	Always H-SUMMARY.
C	Fixed full	Always H-FULL; no joint pre-admission.
D	I-QPG	Per-probe queue-penalized selection; no joint admission.
E	C-QPG	Queue-penalized requests plus joint shared-capacity admission.
F	S-CQPG	Local safety filtering followed by the same joint admission.

F. Statistical Reporting

Action composition, core-required fractions, exact byte totals, analytical queue growth, and admission counts are deterministic under a frozen plan and are reported exactly without confidence intervals. Timing-derived quantities vary across physical sessions; the session is the independent replicate ($n = 12$). Paired session-bootstrap intervals are used only as descriptive timing contrasts. For the separately executed CRFA extension, matching is by frozen session/workload identity rather than concurrent wall-clock execution.

G. Offline Sensitivity and Conformance Analyses

Five read-only analyses complement the hardware campaign without modifying its accepted raw evidence.

Queue-weight sensitivity. The exact frozen I-QPG objective and allocator are replayed over 12 formal seeds for

$$v \in \{1, 3, 10, 30, 100, 168, 300, 1000, 3000, 10000, 30000, 100000\}.$$

The $v = 1$ replay must reproduce the hardware action mix, payload, core fraction, and sampled maximum queue before the sweep is accepted.

Phase-randomized independent control. Twenty fixed seeds assign each probe an independent initial phase in $[0, 1)$ s. Each probe observes only its own queue; no action/queue exchange and no joint admission are introduced. This is a discrete-event software sensitivity, not a hardware measurement.

Schedule-anchor reconstruction. Both (19) and (20) are reconstructed directly from accepted hardware timestamps for all 648 session/method/capacity/device cells.

Probe-count conformance. A discrete-event simulator evaluates $n \in \{3, 8, 16, 32\}$ and $\rho \in \{1/3, 1/2, 2/3, 3/4, 1\}$ for the summary-preserving $m = n$ case. This tests the integer boundary only; it is not a 32-node hardware experiment.

Rotating-frontier conformance. Using the same 1-ms service accounting as the phase study, the cyclic construction in Proposition 5 is checked at all three capacities. This software check verifies the action/count semantics independently of the later CRFA hardware extension.

V. RESULTS

A. RQ1: Universal Per-Round Coverage Has a Quantifiable Byte Cost

For the physical testbed, $n = 3$, $B = 128$ KiB, and $S = 16$ KiB, so $\sigma = 1/8$. Figure 1 shows the relaxed byte-budget

TABLE IV

ONE-ROUND FULL-EVIDENCE BYTE FRONTIER AND PRICE OF UNIVERSAL PER-ROUND REMOTE COVERAGE.

Regime	$K_{\text{full}}^{\max}$	$K_{\text{all}}^{\max}$	Price	$f_{\text{byte}}^{\max}$
Low	1	0	1	0.333
Mid	2	1	1	0.667
High	3	3	0	1.000

boundary. When all probes retain remote evidence ($m = n$), the coordinated policies occupy the exact 0/1/3-full operating points predicted by Theorem 1 in every primary-campaign hardware session.

Corollary 1 gives the unrestricted comparison in Table IV and Fig. 2. If non-selected probes may send no remote evidence, the one-round byte frontier is 1/2/3 full probes at low/mid/high capacity. Requiring all three probes to remain remotely represented reduces this to 0/1/3. Universal per-round coverage therefore costs one full-evidence slot at the two binding capacities and no slot at high capacity.

The separately frozen CRFA extension provides a constructive physical test of the unrestricted operating points. In every measured service round, the admitted action pattern is exactly one, two, or three H-FULL actions at low, mid, and high capacity, with H-NONE on non-selected probes. The aggregate measured application payload therefore equals the corresponding 131,072/262,144/393,216-B budget in every measured round; all pre-service queues are zero and no shared-cap violation occurs.

The more informative result is how much of the byte frontier survives the 1-s post-release service SLO. Table V and Fig. 3 report timely full-core results on the common generated-event denominator. CRFA achieves 0.3056, 0.6327, and 0.9660 at low, mid, and high capacity, equal to 91.7%, 94.9%, and 96.6% of the unrestricted byte frontier. All selected full-evidence events eventually return a core result; the remaining gap is therefore deadline lateness, not missing evidence.

At mid capacity, CRFA produces 2,050 timely full-core results from 3,240 generated events, versus 952 under the matched C-QPG sessions. The paired difference is $+0.3389$ with 95% session-bootstrap interval $[0.3306, 0.3478]$. At low capacity, C-QPG deliberately sends summaries only, whereas CRFA yields 0.3056 timely full results per generated event. At high capacity, both methods admit full evidence for every event; their paired mean difference is small and the descriptive interval spans zero: CRFA–C-QPG is -0.0037 with interval

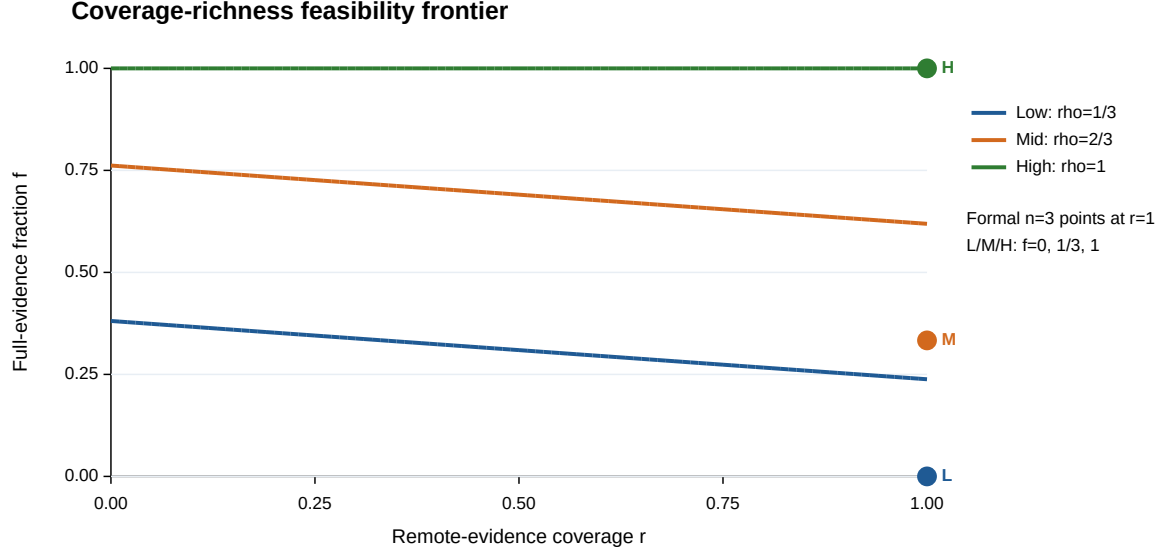


Fig. 1. Relaxed coverage–richness byte-budget lines; admissible points also satisfy $0 \leq f \leq r \leq 1$. Markers are the physical $n = 3$ summary-preserving coordinated operating points.

TABLE V

CRFA HARDWARE ATTAINMENT OF THE UNRESTRICTED FRONTIER. TIMELY FULL IS ON-TIME FULL-CORE RESULTS DIVIDED BY ALL GENERATED EVENTS. C-QPG IS THE MATCHED SUMMARY-PRESERVING HARDWARE COMPARATOR. Δ IS CRFA–C-QPG; INTERVALS ARE DESCRIPTIVE 95% PAIRED SESSION-BOOTSTRAP INTERVALS OVER THE 12 FROZEN SESSION/WORKLOAD IDENTITIES.

Capacity	Byte frontier	C-QPG	CRFA	Attainment	Δ CRFA–C-QPG (95% CI)
Low	0.3333	0.0000	0.3056	0.9167	+0.3056 [+0.3015, +0.3093]
Mid	0.6667	0.2938	0.6327	0.9491	+0.3389 [+0.3306, +0.3478]
High	1.0000	0.9698	0.9660	0.9660	-0.0037 [-0.0133, +0.0046]

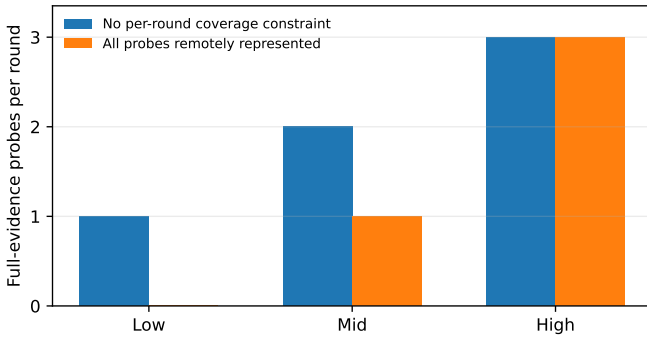


Fig. 2. Maximum one-round full-evidence count with and without the constraint that every probe remains remotely represented.

$[-0.0133, 0.0046]$.

These comparisons do not imply that CRFA dominates C-QPG. CRFA obtains the extra full-evidence slots by assigning no remote evidence to the non-selected probes in a given round. The primary result is therefore a measured price of universal per-round coverage, not a generic ranking of controllers.

a) *Fixed-full overload validation.*: For hardware method C, Proposition 2 predicts per-probe queue growth over 95 generation rounds. The low- and mid-capacity predictions are 7.9167 and 3.9583 MiB, while sampled hardware maxima

are approximately 7.9167 and 3.9584 MiB. At high capacity, $C = 3B$ and deterministic overload disappears. The hardware campaign therefore validates both the feasible side of the byte boundary and the linear-growth consequence of violating the long-run load condition.

B. RQ2: Synchronization, Rate Balance, and Timely-Full Efficiency

At mid capacity, every physical I-QPG arm contains exactly 135 H-FULL and 135 H-NONE measured actions. The core-required fraction is therefore $1/2$, and none of those full requests meets the 1-s post-release SLO. The pattern in Fig. 4 follows Proposition 3: a synchronized full round leaves about 43,691 B of queue per probe, far above the largest positive-payload $v = 1$ threshold (about 260.4 B), forcing a synchronized zero-payload round before the state resets.

The 12-value sweep in Fig. 5 reproduces all 36 physical D session-capacity cells at $v = 1$. The mid-capacity core-required fraction remains exactly $1/2$ through $v = 30,000$ and becomes 0.6222 only at $v = 100,000$; high capacity remains full evidence at every tested v . Low-capacity action composition changes with v , so no all- v cycle identity is claimed.

Phase randomization changes only the probes' initial phases. In this software study, the desynchronized controller uses only H-FULL and H-NONE, and its full-request fractions are exactly

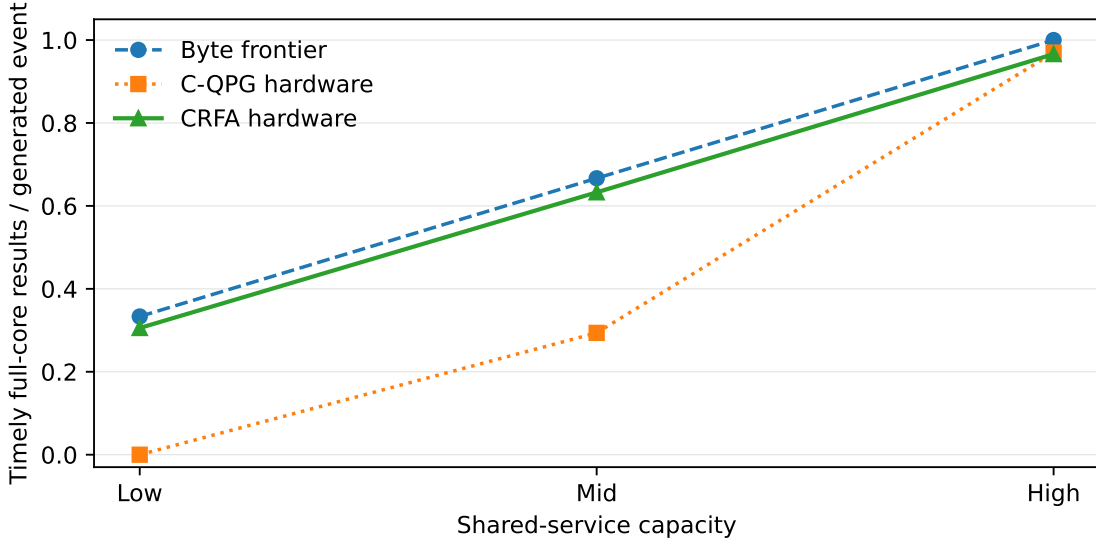


Fig. 3. Physical timely full-core output relative to the unrestricted one-round byte frontier. CRFA approaches the byte frontier at all three capacities, while C-QPG intentionally reserves capacity for per-round remote coverage at low and mid capacity.

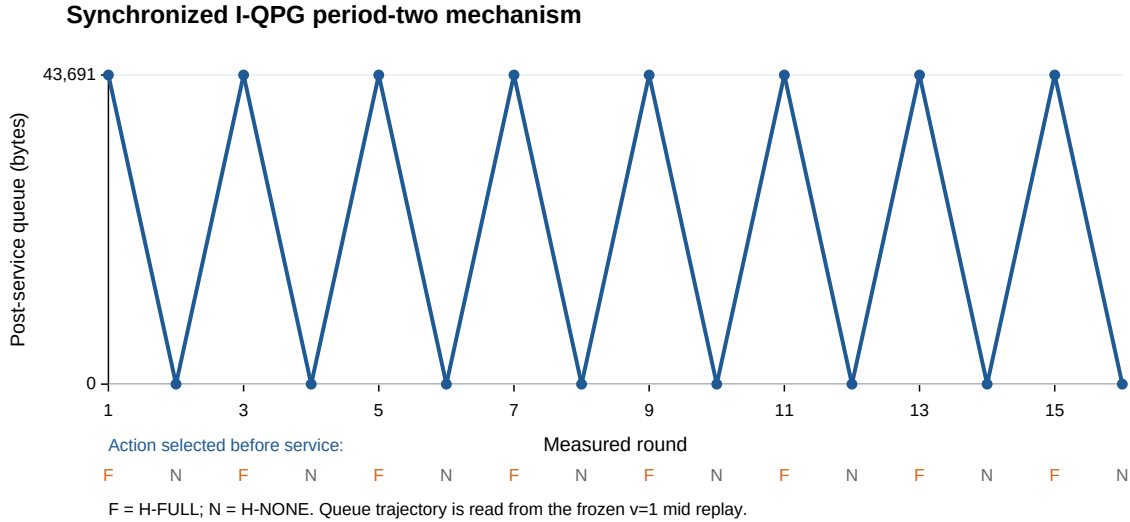


Fig. 4. Frozen $v = 1$ mid-capacity I-QPG queue trajectory and period-two action state.

1/3, 2/3, and 1 at low, mid, and high capacity. These values match Proposition 4 with $p^* = \rho$, providing conformance to the saturated reduced-regime prediction rather than evidence that every decentralized controller has that rate.

The common-denominator mid-capacity software comparison is still useful because it separates request rate from deadline attainment. Table VI divides timely full-core results by all 270 generated events per simulated arm. C-QPG and the desynchronized independent controller produce 0.3333 and 0.3352 timely full results per generated event, respectively, yet C-QPG generates only 14.746 MB versus 23.593 MB—a 37.5% reduction in application bytes. The synchronized controller generates 17.695 MB but produces no timely full-core result.

The new hardware extension changes the interpretation

TABLE VI
MID-CAPACITY SOFTWARE SENSITIVITY ON A COMMON 270-EVENT DENOMINATOR. “TIMELY FULL” IS ON-TIME FULL-CORE RESULTS DIVIDED BY ALL GENERATED EVENTS.

Controller	Full req.	Timely/gen.	Cond.	MB
C-QPG	0.333	0.3333	1.000	14.746
Desync. I-QPG	0.667	0.3352	0.503	23.593
Sync. I-QPG	0.500	0.0000	0.000	17.695

of that counterfactual. At mid capacity, both the phase-randomized independent software policy and CRFA use a 2/3 full-request/admission fraction, but their timely output is very different: 0.3352 in the software phase study versus

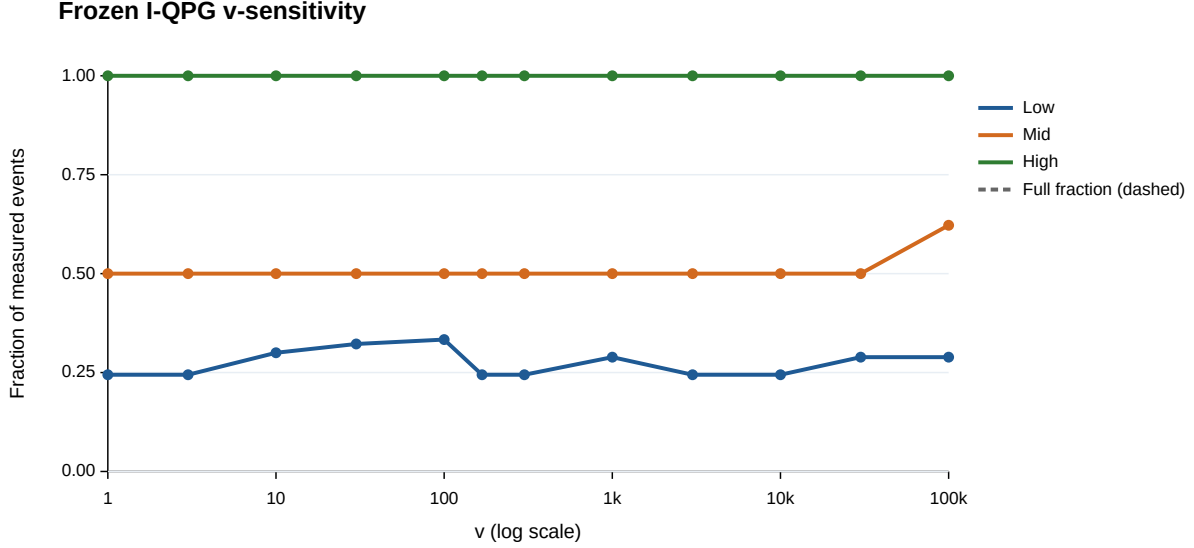


Fig. 5. Frozen-selector sensitivity across the 12 tested queue weights.

0.6327 in the CRFA hardware extension. This is not a direct hardware-versus-hardware method comparison, because the phase-randomized policy remains a software sensitivity. It nevertheless demonstrates why request fraction alone is not a service result: deterministic joint admission can place work directly on the same-round feasible frontier, whereas independent queue dynamics can consume the same long-run byte rate while losing work to deadlines.

C-QPG represents a third operating point. It intentionally gives up one full slot at mid capacity in exchange for a summary from every probe in every round. The relevant distinction is therefore among synchronization, deadline loss, and the security policy of per-round coverage—not a generic centralized-versus-decentralized ranking.

As a first-order occupancy benchmark, if three probes independently request full evidence with probability $p = 2/3$ in a synchronous slot, then $P(J = 1) = 6/27$, $P(J = 2) = 12/27$, and $P(J = 3) = 8/27$. With a $2B$ service budget, all requests complete within a round for $J \leq 2$ but none does for $J = 3$ under equal sharing, giving conditional timely fraction $5/9 \approx 0.556$. The measured software mean is 0.5028. We treat $5/9$ as an independence approximation, not a proven upper bound; the gap shows that queue-state coupling and residual service history matter.

C. RQ3: Physical Hardware Adds Timing Residuals Beyond Byte Feasibility

The analytical region predicts which payload tuples fit the application budget; it does not predict OS scheduling, barrier waiting, Wi-Fi timing, serialization, core execution, or result-return delay.

At high capacity, C, D, E, and F all use H-FULL for every measured event. Mean post-release on-time core-result coverage is 0.9636, 0.9593, 0.9698, and 0.9651, respectively.

The paired F–E mean difference is -0.00463 with a descriptive 95% session-bootstrap interval $[-0.01327, 0.00278]$; the F–D interval also includes zero. Thus, once the byte constraint is non-binding, the primary hardware data do not show a timing advantage for the safety-filtered method.

At mid capacity, E and F each admit one full request per three generated events. Conditional post-release on-time rates among those core-required events are about 0.8815 and 0.8935. One-round byte feasibility therefore does not imply perfect physical SLO attainment.

a) *Frontier attenuation in the rotating extension.*: CRFA makes that residual directly measurable because its admitted full counts equal the analytical unrestricted frontier. Every selected full event eventually returns a core result at all three capacities, yet the conditional post-release on-time rates are 0.9167, 0.9491, and 0.9660. Equivalently, the physical system realizes 91.7%, 94.9%, and 96.6% of the low/mid/high byte frontier within the 1-s service SLO. Admission opportunities are exactly balanced across devices: over the 12 sessions each probe receives 360, 720, and 1,080 full opportunities at low, mid, and high capacity, with maximum inter-selection intervals of three, two, and one rounds. Timing remains device-dependent, however. At mid capacity, conditional post-release rates are 0.9153, 0.9319, and 1.0000 for P1, P2, and P3, respectively. Thus, equal admission opportunity does not imply identical physical deadline attainment.

b) *Schedule-relative release offset.*: Table VII and Fig. 6 aggregate primary-campaign core-required occurrences by physical probe. P1 and P2 have zero nominal-anchor on-time coverage across these occurrences, even though their post-release service coverage is nonzero; P3 retains 0.630 under both anchors. P1’s median release offset alone is about 1.97 s, already larger than the 1-s SLO.

The rotating extension reproduces the same anchor asymmetry. P1 and P2 contribute no nominal-on-time full results,

TABLE VII
PRIMARY-CAMPAIGN CORE-REQUIRED TIMING BY PROBE. COVERAGE IS
WEIGHTED BY CORE-REQUIRED EVENT COUNT.

Probe	Post	Nominal	Lateness p50 (ms)
P1	0.583	0.000	1972.3
P2	0.588	0.000	763.9
P3	0.630	0.630	83.6

TABLE VIII
ANALYTICAL MAXIMUM FULL COUNT k^* FOR $m = n$; ALL 20 VALUES
MATCH THE DISCRETE-EVENT CONFORMANCE SIMULATION.

n	$\rho = 1/3$	$1/2$	$2/3$	$3/4$	1
3	0	1	1	2	3
8	1	3	4	5	8
16	3	6	9	11	16
32	7	13	19	22	32

whereas P3 is nominally on time for essentially every selected full event. Consequently, CRFA nominal timely-full output on the generated-event denominator is only 0.1111, 0.2222, and 0.3330 at low, mid, and high capacity, compared with post-release values of 0.3056, 0.6327, and 0.9660.

Source and log inspection identify a mechanism for persistent phase offset: each probe defines an independent nominal clock before window 0, core submit forms a three-probe barrier, unequal first-iteration waiting appears immediately, and each probe subsequently sleeps relative to its completed iteration rather than catching up to an absolute schedule. The exact OS or network reason that proposals repeatedly reached the barrier in the observed P3/P2/P1 order remains unknown.

Crucially, P1's larger nominal release offset is an orchestration-phase measurement, not evidence that Raspberry Pi 5 computation or inference is slower. It occurs before the post-release service interval. The primary 1-s claims are therefore explicitly limited to a *post-release service SLO*; nominal-anchor results are a schedule-inclusive sensitivity.

D. RQ4: Scaling Conforms to the Integer Boundary and the Hardware Result Chain Is Reconstructable

For the summary-preserving case,

$$f_n^*(\rho) = \frac{1}{n} \left[\left[n \frac{\rho - \sigma}{1 - \sigma} \right] \right]_0^n, \quad \sigma = \frac{1}{8}.$$

A discrete-event conformance study evaluates $n \in \{3, 8, 16, 32\}$ and five normalized capacities. All 20 simulated full counts match the analytical integer result exactly (Table VIII, Fig. 7). This is a conformance check, not physical 32-probe scalability evidence.

a) *Raw reconstruction.*: The primary full-raw audit rebuilds all 216 accepted hardware arms from 58,320 measured occurrences with zero raw-to-validator/event-metric mismatch and zero shared-cap violation across 63,720 service rounds. Its verdict remains `PASS_WITH_ISSUES` because two archival gaps are disclosed: a direct S03-a01 SSD-failure artifact and S07 pre-hardware launcher-abort files are absent from the raw bundle. Neither gap changes an accepted arm.

The separately frozen CRFA extension reconstructs 36/36 accepted arms, 9,720 measured occurrences, and 10,620 shared-service rounds. All event-identity, action/payload-identity, final-drain, reconstruction, and cap gates pass, with zero shared-cap violations. Its attempt index preserves all 55 attempts, including 19 infrastructure exclusions; timing outcomes were not used as acceptance gates. Across the two hardware campaigns, 252 accepted arms, 68,040 measured occurrences, and 74,340 reconstructed service rounds are accounted for, with no shared application-byte cap violation.

The read-only offline analyses contain 432 queue-weight rows, 240 phase-study rows, 648 primary-campaign timestamp-reconstruction rows, and 20 probe-count conformance rows, with no retained failed analysis. Only timestamp reconstruction is derived from measured hardware timestamps; the queue-weight, phase, and probe-count studies are replay/simulation evidence. The CRFA timing reconstruction is separately derived from its 9,720 accepted hardware occurrences.

b) *Role of the safety filter.*: The safety-filtered method records 3,240 H-FULL→H-SUMMARY local safety interventions at low capacity, so the path is active. However, ordinary coordinated admission reaches the same final low-capacity summary mix. At mid capacity, E and F each record 2,160 full-to-summary capacity admission degradations, and at high capacity neither mechanism degrades full evidence. The data support the filter as an auditable local-feasibility interface, not as an independently superior performance mechanism.

c) *Classifier correctness.*: The formal feature replay yields perfect classification on the scored occurrences for all frozen detector paths, so correctness adds no information to the service metric. The dataset is CICIOT2023-derived [5]; the formal occurrence set excludes Recon-PortScan, and exact feature-vector overlap exists across frozen partitions. In keeping with long-standing cautions about closed-world IDS evaluation [6] and recent open-set IIoT work [7], we treat detector correctness as a controlled condition rather than evidence of open-world IDS superiority.

VI. DISCUSSION

A. The Central Trade-off Is Per-Round Coverage versus Timely Richness

The hardware extension makes the coverage price operational. At mid capacity, summary-preserving C-QPG spends $2S = 32$ KiB on two summaries whenever one probe receives full evidence. Its timely full-core output is 0.2938 per generated event. Relaxing same-round universal coverage lets CRFA admit two full uploads and produce 0.6327. The paired increase, 0.3389 with descriptive interval $[0.3306, 0.3478]$, characterizes this coverage–richness exchange under the tested workload, not a causal A/B effect.

At low capacity, C-QPG retains universal coarse visibility but no full-evidence output, whereas CRFA produces 0.3056 timely full results per generated event. At high capacity, both can admit all three full uploads and the paired timing interval spans zero. Rotation thus exchanges continuous coarse visibility for richer evidence over time; neither operating point is universally preferable.

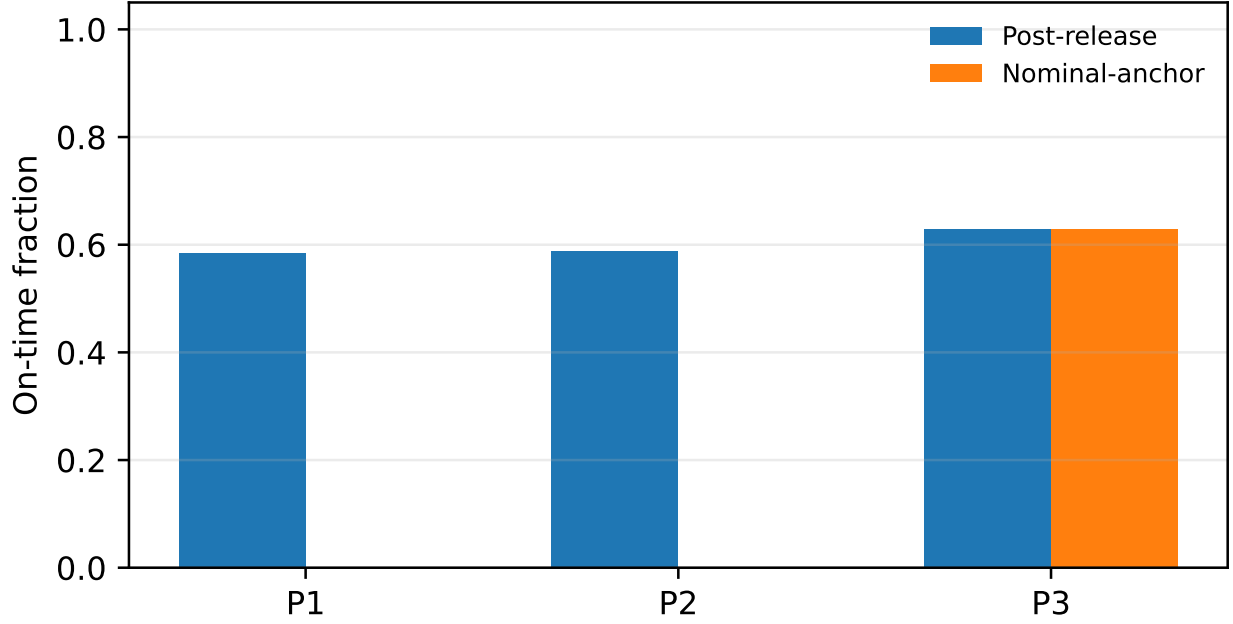


Fig. 6. Primary-campaign post-release versus nominal-anchor on-time coverage, aggregated over core-required occurrences by probe.

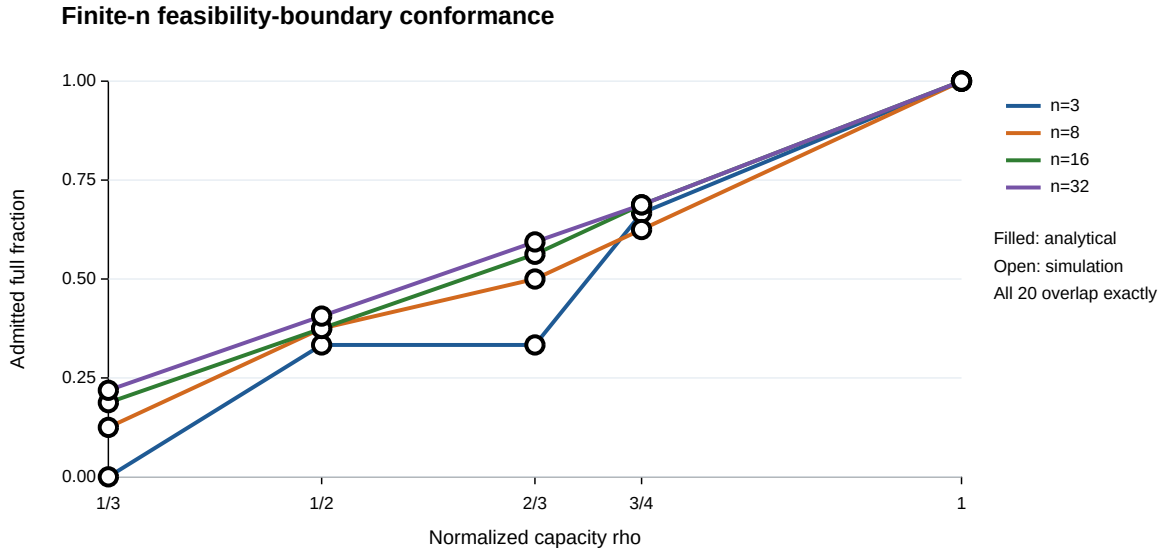


Fig. 7. Discrete-event conformance to the analytical finite- n summary-preserving boundary.

B. Synchronization and Admission Are Distinct Mechanisms

At mid capacity, phase-randomized I-QPG requests full evidence for $2/3$ of generated events, matching the fraction admitted by CRFA. Yet its software timely-full output is 0.3352, versus 0.6327 for CRFA on hardware. This is an interpretive contrast, not a direct method comparison across matched physical conditions. It shows why full-request rate alone cannot determine deadline-level service: independent queue dynamics can use the admissible long-run rate while retaining deadline misses.

Synchronized I-QPG additionally suffers phase locking,

which creates the mid-capacity full/none orbit. Coverage reservation, deadline loss, and synchronization are distinct mechanisms, not a centralized-versus-decentralized ranking.

C. Byte Feasibility Is Necessary but Not Sufficient

The hardware campaigns measure what byte equations cannot: whether a feasible upload returns a core result within the service horizon. Every selected CRFA full result eventually returns, but only 91.7%, 94.9%, and 96.6% of the low/mid/high byte frontier is realized within the post-release SLO. The residual is timing, not selected-evidence loss.

Identical opportunity counts likewise do not ensure identical timing. P3 is essentially always on time after release, whereas P1 and P2 have lower conditional rates. The primary campaign's schedule-relative phase offset persists in the extension, separating admission fairness from end-to-end service fairness.

D. Practical Implications for Deployment

Choose the coverage requirement before the controller. If every probe must retain remote evidence each round, reserve the summary baseline ($C \geq nS$) and admit full upgrades within the remaining budget. If intermittent remote coverage is acceptable, CRFA exchanges per-round universality for full-evidence opportunities with measured maximum inter-selection intervals of 3/2/1 rounds at low/mid/high capacity. These are admission-opportunity bounds, not guarantees on result-return time or evidence age.

Supplementary Section IV derives two planning extensions: a binomial-quantile budget for a tolerated probability of any full-to-summary downgrade under independent within-round requests, and a summary-baseline/upgrade-cost formulation for heterogeneous payloads. They are analytical planning models, not additional hardware findings or models fitted to QPG request traces.

Without a coordinator, phase offsets remain a candidate mitigation for lockstep decisions, not a guarantee of byte-frontier attainment. For schedule-inclusive deadlines, evaluate absolute-clock release scheduling through aligned reruns of both campaigns; the post-release SLO alone does not certify this stricter requirement.

E. Evidence Richness Is Not Security Utility

A summary and a full upload provide different downstream information. Reducing bytes may improve timeliness while reducing forensic richness; additional full evidence may increase deadline pressure. The framework does not map evidence bytes to incident-resolution utility or establish a security-optimal policy. Operators must choose the coverage–richness requirement within the measured service-feasibility boundary.

VII. LIMITATIONS

Three-probe physical scale. The hardware testbed contains three probes. Multi- n results are analytical and discrete-event conformance checks, not physical validation at 8, 16, or 32 nodes.

Rotating extension scope. The CRFA extension physically validates one deterministic cyclic construction at $n = 3$; it does not establish optimality over all admission policies, traffic patterns, or security objectives. Its comparisons with the primary campaign are matched by frozen session/workload identity but were not executed concurrently, so paired intervals are descriptive rather than causal A/B estimates. To preserve session-matched comparability, the extension intentionally inherited the accepted campaign's service accounting, event schema, timestamp fields, scoring semantics, and orchestration timing rather than modifying the launcher; nominal-anchor results therefore remain a schedule-inclusive sensitivity, and

absolute-clock-aligned reruns of both campaigns are left to future work.

Phase randomization is a software sensitivity. The desynchronized independent baseline is not a new Pi experiment. It shows that synchronized failure is not intrinsic to decentralization, but it does not measure Wi-Fi timing for a physical phase-randomized deployment.

Application budget versus physical link rate. Low/mid/high are application-byte service budgets per coordination round, not measured PHY throughput.

Timing anchor. The primary 1-s metric starts at actual event creation. Schedule-inclusive nominal anchoring is materially stricter: P1 and P2 have zero aggregated nominal-anchor on-time coverage across the core-required hardware occurrences. The identified mechanism explains persistence of phase offset, but the exact OS/network cause of the recurring proposal-arrival order remains unknown.

Replay and generalization scope. The workload is a frozen CICIOT2023-derived feature replay [5], not an uncontrolled packet-level deployment. Recon-PortScan, a major validation-error family, is absent from the formal replay, and exact feature-vector repetition exists across frozen partitions. The perfect formal occurrence-level classification therefore does not imply open-world IDS generalization [6], [7].

Shield separation. The tested campaign does not contain a case where a local device constraint rejects an action that is independently feasible under the shared network budget. The safety filter is therefore an implementation/audit interface in this paper, not an independently proven performance layer.

Evidence value. The experiment measures coverage, richness, byte demand, and timeliness, not the downstream incident-resolution value of a summary versus full evidence. Future work should connect the feasibility region to measured forensic or cross-probe correlation utility.

VIII. REPRODUCIBILITY AND PROVENANCE

The primary hardware campaign uses protocol NSF12_20260906_RC1. Accepted scientific attempts are S01-a01, S02-a01, S03-a02, S04-a02, S05-a01, S06-a01, S07-a02, and S08–S12-a01. S03-a01, S04-a01, and S07-a01 are invalid infrastructure attempts and are excluded in full rather than stitched arm-by-arm. The primary full-raw audit remains PASS_WITH_ISSUES only for the two disclosed archival gaps described in Section V.

The rotating extension uses the separately frozen protocol NSF12_CRFA_20260913. It contains 36 accepted arms from 55 indexed attempts; 19 infrastructure failures are retained in the attempt index and excluded in full. Timing outcomes were not acceptance gates. The extension reconstructs all 9,720 measured occurrences and 10,620 service rounds, with all event/action/payload/drain/cap gates passing. Frozen markers for the original 216-arm campaign were checked unchanged before and after the extension.

The primary campaign, extension, raw audits, and read-only offline analyses are hashed separately. Offline sensitivity and conformance analyses do not modify accepted hardware evidence. The artifact package records that the CRFA mirror

resides on the same user-authorized external physical volume as its primary copy and is therefore an integrity-checked mirror rather than an independent physical backup. Before public release, the final artifact will expose persistent checksums and a durable archival identifier. Dataset redistribution rights are handled separately; reproducibility does not require redistributing third-party CICIOT2023 content [5].

IX. CONCLUSION

NETSHIELD characterizes evidence service under a shared bottleneck through the one-round region $kB + (m - k)S \leq C$, its unrestricted full-evidence frontier, and distinct finite-deadline and long-run load conditions. Requiring universal per-round remote coverage costs one full-evidence slot at low and mid capacity in the three-probe system, but none at high capacity.

The cyclic hardware construction attains 91.7%, 94.9%, and 96.6% of the low/mid/high byte frontier. At mid capacity, timely full-core output rises from 0.2938 under session-matched C-QPG to 0.6327 under CRFA, with paired difference +0.3389 and descriptive 95% session-bootstrap interval [0.3306, 0.3478]. This is a coverage–richness trade-off, not universal controller dominance. Synchronized queue feedback creates a separate full/none limit cycle; the phase-randomized software study conforms to the conditional rate-balance prediction without guaranteeing timely completion.

Every selected CRFA full result is eventually returned, so its gap to the byte frontier is timing rather than selected-evidence loss. Schedule-relative release offsets further separate post-release service from nominal deadlines. Across 252 accepted hardware arms, 68,040 measured events and 74,340 reconstructed service rounds are accounted for with zero shared application-byte cap violations. Together, the analysis, constructive controller, and auditable hardware measurements expose which evidence allocations are feasible and how closely physical service realizes them.

REFERENCES

- [1] S. Raza, L. Wallgren, and T. Voigt, “SVELTE: Real-time intrusion detection in the Internet of Things,” *Ad Hoc Networks*, vol. 11, no. 8, pp. 2661–2674, 2013.
- [2] M. Miettinen, S. Marchal, I. Hafeez, N. Asokan, A.-R. Sadeghi, and S. Tarkoma, “IoT Sentinel: Automated device-type identification for security enforcement in IoT,” in *Proc. IEEE ICDCS*, 2017, pp. 2177–2184.
- [3] Y. Meidan *et al.*, “N-BaIoT—Network-based detection of IoT botnet attacks using deep autoencoders,” *IEEE Pervasive Computing*, vol. 17, no. 3, pp. 12–22, 2018.
- [4] Y. Mirsky, T. Doitshman, Y. Elovici, and A. Shabtai, “Kitsune: An ensemble of autoencoders for online network intrusion detection,” in *Proc. NDSS*, 2018.
- [5] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, “CICIOT2023: A real-time dataset and benchmark for large-scale attacks in IoT environment,” *Sensors*, vol. 23, no. 13, Art. no. 5941, 2023, doi: 10.3390/s23135941.
- [6] R. Sommer and V. Paxson, “Outside the closed world: On using machine learning for network intrusion detection,” in *Proc. IEEE Symp. Security and Privacy*, 2010, pp. 305–316, doi: 10.1109/SP.2010.25.
- [7] X. Yang, F. Tong, F. Jiang, and G. Cheng, “A lightweight and dynamic open-set intrusion detection for Industrial Internet of Things,” *IEEE Trans. Inf. Forensics Security*, vol. 20, pp. 2930–2943, 2025, doi: 10.1109/TIFS.2025.3546849.
- [8] S. M. M. Mirnajafizadeh, A. R. Sethuram, D. Mohaisen, D. Nyang, and R. Jang, “Enhancing network attack detection with distributed and in-network data collection system,” in *Proc. 33rd USENIX Security Symp.*, 2024, pp. 5161–5178.
- [9] M. Oqaily, H. Purohit, Y. Jarraya, L. Wang, B. Nour, M. Pourzandi, and M. Debbabi, “ChainPatrol: Balancing attack detection and classification with performance overhead for service function chains using virtual trailers,” in *Proc. 33rd USENIX Security Symp.*, 2024, pp. 3441–3458.
- [10] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, “A survey on mobile edge computing: The communication perspective,” *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 2017.
- [11] P. Mach and Z. Becvar, “Mobile edge computing: A survey on architecture and computation offloading,” *IEEE Commun. Surveys Tuts.*, vol. 19, no. 3, pp. 1628–1656, 2017.
- [12] Y. Zhao, C. Qiu, X. Shi, X. Wang, D. Niyato, and V. C. M. Leung, “Cur-CoEdge: Curiosity-driven collaborative request scheduling in edge-cloud systems,” in *Proc. IEEE INFOCOM*, 2024, pp. 411–420, doi: 10.1109/INFOCOM52122.2024.10621190.
- [13] Y. Hao, S. Yang, F. Li, Y. Zhang, S. Wang, and X. Ren, “EdgeTimer: Adaptive multi-timescale scheduling in mobile edge computing with deep reinforcement learning,” in *Proc. IEEE INFOCOM*, 2024, pp. 671–680, doi: 10.1109/INFOCOM52122.2024.10621305.
- [14] F. Li, Y. Hao, S. Yang, and P. Zhao, “OACR²: Online admission control and resource reservation for 5G slice networks with deep reinforcement learning,” *IEEE Trans. Mobile Comput.*, vol. 24, no. 8, pp. 7360–7376, 2025, doi: 10.1109/TMC.2025.3548767.
- [15] A. Demers, S. Keshav, and S. Shenker, “Analysis and simulation of a fair queueing algorithm,” in *Proc. ACM SIGCOMM*, 1989, pp. 1–12.
- [16] A. K. Parekh and R. G. Gallager, “A generalized processor sharing approach to flow control in integrated services networks: The single-node case,” *IEEE/ACM Trans. Netw.*, vol. 1, no. 3, pp. 344–357, 1993.
- [17] L. Tassiulas and A. Ephremides, “Stability properties of constrained queueing systems and scheduling policies for maximum throughput in multihop radio networks,” *IEEE Trans. Autom. Control*, vol. 37, no. 12, pp. 1936–1948, 1992.
- [18] M. J. Neely, *Stochastic Network Optimization with Application to Communication and Queueing Systems*. Morgan & Claypool, 2010.
- [19] J. Huang, L. Golubchik, and L. Huang, “When Lyapunov drift based queue scheduling meets adversarial bandit learning,” *IEEE/ACM Trans. Netw.*, vol. 32, no. 4, pp. 3034–3044, 2024, doi: 10.1109/TNET.2024.3374755.
- [20] D. Crankshaw, X. Wang, G. Zhou, M. J. Franklin, J. E. Gonzalez, and I. Stoica, “Clipper: A low-latency online prediction serving system,” in *Proc. 14th USENIX NSDI*, 2017, pp. 613–627.
- [21] H. Shen, L. Chen, Y. Jin, L. Zhao, B. Kong, M. Philipose, A. Krishnamurthy, and R. Sundaram, “Nexus: A GPU cluster engine for accelerating DNN-based video analysis,” in *Proc. 27th ACM SOSP*, 2019, pp. 322–337.
- [22] A. Gujarati, R. Karimi, S. Alzayat, W. Hao, A. Kaufmann, Y. Vigfusson, and J. Mace, “Serving DNNs like Clockwork: Performance predictability from the bottom up,” in *Proc. 14th USENIX OSDI*, 2020, pp. 443–462.
- [23] D. Crankshaw, G.-E. Sela, X. Mo, C. Zumar, I. Stoica, J. E. Gonzalez, and A. Tumanov, “InferLine: Latency-aware provisioning and scaling for prediction serving pipelines,” in *Proc. ACM SoCC*, 2020, pp. 477–491.
- [24] A. Khare, D. Garg, S. Kalra, S. Grandhi, I. Stoica, and A. Tumanov, “SuperServe: Fine-grained inference serving for unpredictable workloads,” in *Proc. 22nd USENIX NSDI*, 2025, pp. 739–758.
- [25] M. K. C. Shisher, Y. Sun, and I.-H. Hou, “Timely communications for remote inference,” *IEEE/ACM Trans. Netw.*, vol. 32, no. 5, pp. 3824–3839, 2024, doi: 10.1109/TNET.2024.3408673.
- [26] C. Ari, M. K. C. Shisher, Y. Sun, and E. Uysal, “Goal-oriented status updating for real-time remote inference over networks with two-way delay,” *IEEE Trans. Netw.*, vol. 34, pp. 4011–4025, 2026, doi: 10.1109/TNET.2026.3670099.
- [27] M. Alshiekh *et al.*, “Safe reinforcement learning via shielding,” in *Proc. AAAI*, 2018, pp. 2669–2678.
- [28] K. P. Wabersich and M. N. Zeilinger, “A predictive safety filter for learning-based control of constrained nonlinear dynamical systems,” *Automatica*, vol. 129, Art. no. 109597, 2021.
- [29] N. Ouyang, A. Shatte, Z. Lu, C. Chen, and W. Xiang, “SoK: Telemetry-aware runtime assurance for always-on on-device intrusion detection,” in *Information Security and Privacy (ACISP 2026)*, LNCS 16794, Springer, 2026, pp. 163–183.